

Nandini Chinta

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Summary

- Data Engineer & Interim Technical Lead with around **7 years of experience** delivering scalable, enterprise-grade data platforms, dbt semantic layers, and curated analytical datasets across GCP, AWS, and Azure.
- Proven track record establishing technical strategy, dbt data modeling patterns, metric governance standards, and real-time event streaming pipelines.
- Expertise in optimizing query performance (**40%**), establishing data quality frameworks (**99.8% accuracy**), and managing high-concurrency analytical workloads (**100+ TB/day**).
- Hands-on experience deploying ML systems, LLM-powered applications, and privacy-aware AI features in production.
- *Certified Google Cloud Professional Data Engineer*

Skills

Languages: Python, SQL, Java

Data Engineering & Modeling: dbt, BigQuery, Dataflow, Datastream, Dimensional & Canonical Data Modeling, Semantic Layers, Metric Governance, ETL/ELT, Streaming & Batch Pipelines

Data Governance & Observability: Data Lineage, Anomaly Detection, Row-Level Security, Access Governance, Dataplex, SLA Tracking

Cloud & Infrastructure: GCP (BigQuery, Pub/Sub, Cloud Run, Cloud SQL, Document AI), AWS (SageMaker), Azure (Data Factory, Synapse)

Systems & Tools: FastAPI, Docker, Terraform, GitLab CI/CD, Prefect, Spring Boot, Git, Jira

Visualization & Analytics: Looker, Tableau, Power BI, Matplotlib

ML & AI Infrastructure: Vertex AI, Feature Stores, Vector Search, RAG Architecture, PyTorch, TensorFlow

Experience

Data Engineer & Interim Technical Lead – Data/AI Platform | Veolia North America, New Jersey

June 2023 - Present

- **Lead and manage a team** of data engineers, defining architectural direction, scoping complex analytical requirements, and setting **SLAs** across 6 enterprise business units.
- Architected and deployed a company-wide **dbt self-service semantic layer** across BigQuery, establishing **single-source-of-truth** canonical metric definitions across business units and reducing metric drift to zero.
- Architected end-to-end **streaming pipelines** using Datastream, Dataflow, BigQuery, and dbt to process **100+ TB/day** of **high-concurrency event telemetry** (user activity, session states, transactional data), reduced data availability latency from days to seconds and query latency by **40%**.
- Partnered with Product, Finance, and Engineering stakeholders to surface **performant analytical datasets** in Looker and PowerBI, delivering **executive-level** dashboards with sub-5 minute latency.
- Standardized ingestion architectures spanning CDC, API, SFTP, and batch integrations, enabling one-click onboarding of 150+ data sources into BigQuery.
- Implemented **enterprise data governance frameworks** processing sensitive HR and audit datasets using IAM, row-level security, taxonomy management, and privileged access management (PAM) principles.
- Built a **self-service Access Management Platform** using FastAPI and Pydantic, enabling policy-based provisioning and automated access lifecycle tracking, reducing turnaround time from 2 days to under 2 seconds.
- Established **data quality standards** using **Dataplex** and **automated dbt testing frameworks**, implementing anomaly detection, schema enforcement, and freshness SLAs across datalakes, raising dataset **accuracy** to **99.8%**.
- Built an **observability framework** incorporating freshness monitoring, SLA tracking, and cost attribution, reducing incident resolution time by **50%** and cloud spend by up to **80%**.
- Developed enterprise **RAG** solutions integrating internal knowledge repositories, vector search, and LLMs to improve document discovery and **semantic search**.
- Designed distributed backend services handling **5K+** concurrent requests (<200ms latency) and deployed distributed ML pipelines on Vertex AI, cutting training time by 30%.
- Architected an on-device TFLite wake-word voice system with noise suppression, buffering audio for Whisper STT and routing queries to a secure internal AI agent.
- Standardized **dbt modeling patterns**, naming conventions, and **ingestion templates** across the organization, mentoring junior engineers to accelerate delivery and harden data assets.

Machine Learning Researcher - SUNY Buffalo, New York

Feb 2022 - June 2023

- **Fairness in Multi-Agent Reinforcement Learning:** Researched fairness in multi-agent systems by developing algorithms that balance individual and collective rewards under constrained environments. Conducted experiments in a custom grid-world, evaluating Q-Learning, DQN, DDQN, and PPO, demonstrating that DDQN and PPO achieve superior coordination, equitable outcomes, and efficient goal completion.
- **Traffic Flow Control using Multi-Agent Reinforcement Learning (MARL):** Researched traffic flow optimization in smart cities by modeling traffic lights as agents and learning policies to maximize overall traffic efficiency. Implemented and evaluated Deep RL algorithms, including Actor-Critic and PPO, using the SUMO simulator, achieving improved coordination and reduced congestion.

Software Engineer - Kaiser Permanente(TCS), India

June 2019 - Aug 2021

Contributed to enterprise healthcare platforms at Kaiser Permanente, building production-grade ML systems, backend services, and data pipelines supporting clinical operations and physician workflows.

- Developed **batch job Microservices**, eliminating **90%** of manual operational effort automating SOP workflows.
- Optimized SQL **queries, stored procedures, and views** for high-volume **transactional** healthcare systems, saving **100+** engineering hours per month in downstream reporting and processing.
- Implemented **concurrency and multithreading** strategies to increase service throughput from **500 → 1,100 TPS** (+120%) and reduce batch execution time by **50%**.
- Integrated Python **ML classification models** into production applications for real-time operational decisioning.
- **Led a team** of engineers, code reviews, defined software design standards, and established CI/CD practices across backend services and data workflows.

Research & Projects

RAG Applications - Veolia

Python, GCP, LLMs

- Built enterprise RAG applications integrating **vector search** and **LLMs** for document intelligence, improving query accuracy by **40%** and reducing manual search time.
- Deployed optimized pipelines using **LangChain + GCP**, enabling scalable document ingestion and retrieval while lowering latency.
- Developed **Streamlit-based UI applications** for RAG systems, providing user-friendly interfaces for business teams to query documents and access insights seamlessly.

GenAI with LLMs

Python, AWS Sagemaker, LLMs

- Utilized **data parallelism and Microsoft's ZeRO optimizations** to manage large-scale models, reducing computational costs by 40% without compromising performance.
- Applied Reinforcement Learning from Human Feedback (**RLHF**), increasing model alignment with human values by **50%**, as evidenced by user satisfaction.
- Deployed Retrieval-Augmented Generation (**RAG**), ensuring more precise answers and reducing **hallucination**.

Education

Master of Science in Artificial Intelligence

Aug 2021 – Feb 2023

University at Buffalo, SUNY, Buffalo, NY, CGPA: 3.6/4

- **Teaching Assistant:** Courses: 574 Machine Learning, 546 Reinforcement Learning

B. Tech in Computer Science and Engineering

Jul 2015 – May 2019

Vignan University, India, CGPA: 9.05/10

Certifications

- Google Cloud Professional Data Engineer [[link](#)]
- Terraform Associate [[link](#)]
- Prefect Associate [[link](#)]
- GitLab CICD